
When Uncertainty Sampling Hurts: Scale-Free Space-Filling Acquisition for Kriging-Driven Autonomous Experiments

Anonymous Authors

Abstract

Autonomous experiments that reconstruct an unknown response surface with a Gaussian-process (Kriging) surrogate commonly select each next measurement where the posterior variance is largest—the SMART criterion of Noack et al. (2019) and the default of several autonomous-experimentation platforms. We show that this pure-uncertainty rule has a structural failure mode: whenever the fitted correlation length becomes comparable to the parameter domain—as happens on heterogeneous surfaces—the variance landscape saturates into a plateau whose maxima sit on the domain boundary, and the experiment over-invests in boundary regions while the interior, where the reconstruction error actually lives, stays under-resolved. On a three-dimensional Stokes–Einstein diffusion benchmark spanning four decades in the response, uncertainty sampling is a factor of two *worse* than naive random sampling at matched budgets. We propose a simple, scale-free repair, *boosted-variance sampling* (BVS): the Kriging variance is multiplied by an exponential space-filling factor whose argument is the squared ratio of the distance to the nearest measurement to the median spacing of the current design, making the criterion dimensionless and budget-adaptive. On a five-function benchmark with a common Kriging surrogate, ten seeds, and budgets from 50 to 1000 measurements, BVS dominates the uncertainty-sampling and gradient-weighted baselines at every budget on every function, doubles the measurement efficiency of SMART and GSMART on the diffusion benchmark, and adds at most about a third to the per-step cost (and is slightly faster than SMART on one function). Complementary oracle-design experiments bound what *any* acquisition strategy can achieve under the shared surrogate: on the four two-dimensional test functions the headroom above BVS is at most a few percent, so order-of-magnitude improvements there require better surrogates, not better acquisition. All code, logs, and per-seed records are released for verification.

1 Introduction

A growing family of instruments—synchrotron scattering beamlines, molecular-foundry synthesis robots, automated electron microscopes—can acquire data far faster than humans can direct them [Kusne et al., 2014, MacLeod et al., 2020, Noack et al., 2019a, Yager et al., 2023]. In many of these settings the scientific goal is not to locate a single optimum but to *reconstruct a response surface*: the experimenter wishes to know a scalar quantity $\rho(\mathbf{p})$ —a peak intensity, a structural classification, a transport coefficient—as a function of sample position and environmental parameters $\mathbf{p}$, with each measurement costing seconds to hours of beamtime. Because a full grid scan may require thousands of measurements, the instrument should decide for itself where to measure next, using only the data collected so far.

The most influential such framework for scattering experiments is SMART [Noack et al., 2019a]: an ordinary-Kriging surrogate with an exponential variogram is refit after every measurement, and the

next measurement is placed at the maximum of the Kriging posterior variance. A gradient-weighted variant, GSMART, steers measurements toward high-gradient regions and enabled edge-resolved autonomous imaging [Noack et al., 2020]. The same principle—uncertainty sampling [Settles, 2009, Seo et al., 2000]—appears throughout the active learning and computer-experiments literature as maximum-variance, maximum-entropy, or integrated-mean-squared-error (IMSE) infill criteria [Sacks et al., 1989, Santner et al., 2003, Liu et al., 2017]. When the surrogate is well specified, greedy variance maximization provably reduces the integrated posterior variance, and it works well in practice.

This paper is about what happens when the standard assumptions do not hold. Real response surfaces are heterogeneous: amplitude and local complexity vary across the parameter space by orders of magnitude, and the globally fitted correlation length can grow to the size of the domain itself. We show, on the five synthetic benchmarks introduced by Noack et al. [2019a], that pure uncertainty sampling then develops a structural pathology: with a saturated kernel the Kriging variance is a plateau whose maxima lie on the boundary of the domain, so the experiment concentrates measurements on faces, edges, and corners while the interior—where the reconstruction error mass actually sits—stays under-resolved. On the three-dimensional Stokes–Einstein diffusion benchmark, a physically motivated surface spanning four decades in D , SMART places 47% of its measurements within 5% of the domain boundary (a volume-proportional design places 27%) and ends up a factor of two worse than *random* sampling at every tested budget; the original paper’s three-dimensional advantage over grid scans survives only because the grid baseline suffers the same heterogeneity more severely.

Our contribution is threefold.

1. A scale-free acquisition criterion. We propose boosted-variance sampling (BVS),

$$\alpha(\mathbf{p}) = \sigma^2(\mathbf{p}) \exp\left(\beta \frac{d_1(\mathbf{p})^2}{\bar{d}^2}\right),$$

the product of the Kriging posterior variance σ^2 with an exponential space-filling factor: d_1 is the distance to the nearest measurement, $\bar{d}$ is the median spacing of the current design, and β is a single dimensionless constant ($\beta = 4$ throughout, all functions). The normalization by $\bar{d}$ makes the criterion scale-free and budget-adaptive (it automatically sharpens as the design densifies), and the exponential interpolates smoothly between pure uncertainty sampling ($\beta \rightarrow 0$) and sequential maximin (farthest-point) sampling ($\beta \rightarrow \infty$). Unlike gradient-weighted criteria, BVS contains no dimensional quantities, so its behavior does not depend on the physical units or dynamic range of the data.

2. A rigorous common-surrogate benchmark. We re-evaluate SMART, GSMART, grid, random, and BVS on the five test functions of Noack et al. [2019a] under a modern protocol: one shared Kriging surrogate for every method (isolating the acquisition criterion), ten seeds per configuration with paired initial designs, budgets from 50 to 1000, and predictive-quality metrics beyond point error (RMSE, CRPS, and calibration-coverage curves). The full 21-configuration sweep, all per-step hyperparameter traces, and all per-seed logs are released.

3. An oracle bound on acquisition headroom. Using oracle designs with perfect knowledge of each function’s structure—including straddling point pairs along the known discontinuity lines—we bound what any acquisition strategy could achieve under the shared surrogate. On the four two-dimensional functions the bound is within a few percent of what BVS achieves, so claims of large acquisition-driven speedups on these benchmarks are not attainable in principle; improvement there must come from the surrogate (e.g., anisotropic or non-stationary kernels). On the heterogeneous three-dimensional function the headroom is large, and BVS captures the bulk of it: at 500 measurements it already reaches the error that uncertainty sampling needs 1000 measurements to attain.

We also report negative results. The literal GSMART criterion $\sigma^2(1 + c|\nabla\hat{\rho}|^2)$ with $c = 20$ collapses on every discontinuous benchmark under a global-reconstruction metric: it chases permanently unresolved discontinuities, poisons its own variogram fit, and ends up worse than random sampling; a scale-normalized variant still underperforms plain uncertainty sampling, showing the failure is structural rather than a units accident. These findings delimit where gradient-weighted exploration is appropriate (edge-local imaging, as in the original demonstration) and where it is not (global mapping).

2 Related work

Design and analysis of computer experiments. Kriging-based surrogates and the IMSE perspective originate with Sacks et al. [1989]; the standard modern treatment is Santner et al. [2003]. Infill criteria for *optimization* (expected improvement, EGO; Jones et al., 1998) are complementary to ours: they seek a maximizer, while we (following SMART) seek global reconstruction. Sequential criteria for global fit include the maximum-expected-prediction-error family [Liu et al., 2017] and EIGF-style nearest-data-mismatch criteria [Ben Salem, 2015]; these remain variance- or residual-driven and, as our analysis shows, inherit the amplitude heterogeneity of the variance signal. Information-theoretic sensor placement [Krause et al., 2008] similarly maximizes uncertainty reduction under a stationary prior.

Space-filling designs. Maximin distance designs [Johnson et al., 1990], Latin hypercubes [McKay et al., 1979], and centroidal Voronoi tessellations [Du et al., 1999] are the static counterpart of our space-filling factor; sequential farthest-point sampling is its greedy extreme. Our contribution is not a new design but the *composition* of a model-based uncertainty signal with a model-free space-filling signal in units of the current design’s own spacing, which retains the benefits of each in the regimes where the other fails (Section 6.1).

Autonomous experiments. Noack et al. [2019a] introduced SMART/GSMART for scattering experiments; Noack et al. [2020] developed the gradient-weighted variant operationally, and Noack and Reyes [2023] review the mathematical framework. Closed-loop Bayesian active learning has likewise accelerated materials discovery [Kusne et al., 2020]. On the surrogate side, our variogram fitting follows the classical geostatistical treatment of lag-binned empirical variograms [Cressie, 1993, Cressie and Hawkins, 1980], and response-surface infill criteria more broadly are surveyed by Jones [2001]. The gpCAM/fvGP stack [Noack et al., 2019b] provides the surrogate and optimization machinery used here. Autonomous and self-driving laboratories more broadly are reviewed by MacLeod et al. [2020] and Yager et al. [2023]. Our benchmark deliberately reuses the original paper’s test functions so that claims about acquisition quality are directly comparable to the published record.

3 Methodology

3.1 Problem statement

Let $\rho : \mathcal{X} \subset \mathbb{R}^d \rightarrow \mathbb{R}$ be an unknown function on a box domain, accessible through expensive point evaluations. After t measurements $\mathcal{D}_t = \{(\mathbf{p}_i, \rho_i)\}_{i=1}^t$, an *acquisition criterion* $\alpha(\mathbf{p}; \mathcal{D}_t)$ is maximized over $\mathcal{X}$ (by a global optimizer), and the maximizer is measured next. The objective is to minimize the reconstruction error $\text{IAE}(t) = \mathbb{E}_{\mathbf{p}} |\hat{\rho}_t(\mathbf{p}) - \rho(\mathbf{p})|$ at every t , i.e. to make the error–budget curve converge as fast as possible. This is a sequential experimental-design problem, distinct from Bayesian optimization.

3.2 Shared surrogate: ordinary Kriging with an exponential variogram

All methods in this paper—baselines and proposed alike—share one surrogate, so that only the acquisition criterion differs. Following Noack et al. [2019a], we use ordinary Kriging with the exponential variogram $\gamma(h) = 1 - e^{-ah}$, i.e. a Matérn-1/2 kernel

$$k(\mathbf{p}, \mathbf{p}') = s e^{-a \|\mathbf{p} - \mathbf{p}'\|_2}, \quad (1)$$

implemented on gpCAM’s GP0ptimizer [Noack et al., 2019b]. After every measurement the variogram is refit from the data by least squares, reproducing the paper’s Algorithm 2: all pairwise squared differences $(y_i - y_j)^2$ are binned by lag into an empirical variogram (the “average behavior” shown in the original Fig. 2), normalized to unity, and the single parameter a is fitted by multi-start least squares with the sill fixed to one, as in the original kriging system where the correlation matrix $C_{ij} = 1 - \gamma(\|p_i - p_j\|)$ has unit diagonal. The data-unit variance scale s follows from the normalization constant (the largest binned squared difference). This single fitting procedure is applied identically to every function, method, budget, and step; the fitted values (and their full traces) are logged and reported in Appendix B. Measurements are treated as noiseless; a scale-relative numerical floor $\text{var}_i = (10^{-8} \text{sd}(y))^2$ keeps the Gram matrix conditioned regardless of physical

units (an *absolute* floor silently destroys benchmarks whose values are much smaller than the floor; Appendix A).

3.3 Baseline criteria

SMART [Noack et al., 2019a] selects the next measurement at the maximum of the posterior variance,

$$\alpha_{\text{SMART}}(\mathbf{p}) = \sigma^2(\mathbf{p}). \quad (2)$$

GSMART [Noack et al., 2019a], Eq. (10) of that paper, weights the *unit-sill* variance by the squared gradient of the surrogate,

$$\alpha_{\text{GSMART}}(\mathbf{p}) = \frac{\sigma^2(\mathbf{p})}{s} (1 + c |\nabla \hat{\rho}(\mathbf{p})|^2), \quad c = 20, \quad (3)$$

where s is the variogram sill, $\hat{\rho}$ the posterior mean in data units, and $c = 20$ is the value used in the original demonstration. We evaluate Eq. (3) literally, and additionally a scale-normalized variant (GSMART_n) in which the gradient is divided by the data standard deviation; both are reported because the literal form is dimensionally c -dependent (Section 6.2).

3.4 Boosted-variance sampling

Our criterion multiplies the Kriging variance by an exponential space-filling factor:

$$\boxed{\alpha_{\text{BVS}}(\mathbf{p}) = \sigma^2(\mathbf{p}) \exp\left(\beta \frac{d_1(\mathbf{p})^2}{\bar{d}_t^2}\right)} \quad \begin{aligned} d_1(\mathbf{p}) &= \min_{i \leq t} \|\mathbf{p} - \mathbf{p}_i\|_2, \\ \bar{d}_t &= \text{median}_{i \leq t} \min_{j \neq i} \|\mathbf{p}_i - \mathbf{p}_j\|_2, \end{aligned} \quad (4)$$

with a single constant $\beta = 4$ used for all functions and budgets. Every quantity in Eq. (4) is either dimensionless (β , the ratio $d_1/\bar{d}$) or recomputed from the data at every step (σ^2 from the refit surrogate, d_1 and $\bar{d}$ from the design itself), so the criterion is invariant to rescalings of the domain and of the response—there is nothing to re-tune when the instrument changes units.

Motivation. The Kriging variance is a good *local* uncertainty signal only while it is sensitive to distance from the data. For a Matérn-1/2 kernel the variance saturates at the sill within a correlation length of the nearest measurement, so once the design spacing is comparable to $1/a$ the variance landscape is a plateau: nearly constant over most of the domain, with narrow pits at the data. On such a plateau, greedy variance maximization is driven by the kernel’s global geometry rather than by local coverage (Section 6.1), and the global optimizer returns points at the plateau’s rim—domain corners, faces, and boundary layers—regardless of how well those regions are already covered. The exponential factor converts the plateau back into a well-posed maximin-type landscape: at a point one median-spacing away from all data ($d_1 = \bar{d}$) the variance is boosted by e^β , and at two median spacings by $e^{4\beta}$, so holes in the design dominate the argmax. The boost is *multiplicative*, so the variance retains the tie-breaking role wherever the design is genuinely uniform—and with it the model-based preference for genuinely uncertain regions (this is why BVS beats pure maximin sampling on the three-dimensional benchmark at high budgets; Section 6.3).

Limits and relatives. $\beta \rightarrow 0$ recovers SMART (Eq. 2); $\beta \rightarrow \infty$ recovers sequential maximin [Johnson et al., 1990]. BVS is therefore a one-parameter bridge between the model-based and model-free ends of a family that contains both baselines, with the bridge parameter fixed once, globally. Unlike α -blend infill criteria used in surrogate optimization, which mix exploration with *exploitation* of the posterior mean [Gramacy, 2020, Ch. 6], our blend joins two *exploratory* signals that differ in model dependence. Unlike GSMART (Eq. 3), the criterion uses no derivatives of the surrogate and no dimensional constants, and it never rewards proximity to already-measured structure.

Optimization. α_{BVS} , like the variance itself, has narrow pits at measured points and broad plateaus, so we maximize it with the same population-based global optimizer used for the baselines (gpCAM’s differential-evolution wrapper of Storn and Price [1997]; population 12d, 15 iterations, no polish, identical settings for every method). Each evaluation costs one variance query, one nearest-neighbor query, and one median computation; BVS adds no hyperparameter fitting.

4 Experimental setup

Test functions. We use the five synthetic benchmarks of Noack et al. [2019a]: the piecewise-sinusoidal functions of their Eqs. (12) and (13) on $[0, 50]^2$ (smooth oscillations with steps of 140/160 and 200/360 at $p_0 = 30$ and $\|\mathbf{p}\| = 20$; we call them Piecewise I and II); the smooth oscillatory function of their Eq. (15), $\rho = \sin 2p_0 + \sin 2p_1 + (p_0/5)^2 - (p_1/5)^2$ on $[-10, 10]^2$ (Smooth osc.); the non-differentiable, discontinuous piecewise-linear function of their Eq. (17) on $[0, 50]^2$ (Non-diff.); and the three-dimensional Brownian diffusion coefficient of their Eq. (18), $D = k_B T / (6\pi\mu r)$, with $r \in [1, 100]$ nm, $T \in [0, 100]$ °C, glycerol mass fraction $C_m \in [0, 100]\%$, and the Cheng viscosity model [Cheng, 2008]—a smooth but strongly heterogeneous surface spanning four decades in D (3D diffusion).

Protocol. Every run starts from the same 10 uniformly random initial points within a seed; all methods within a seed share this initial design exactly. Budgets are 50/100/200/500 for the 2D functions and 50/100/200/500/1000 for the 3D function (21 configurations); each is run with 10 independent seeds. The grid baseline uses, at each budget B , its own $\lceil B^{1/d} \rceil^d$ lattice (randomly thinned to B if needed), matching the original protocol of equal axis subdivisions; the random baseline samples uniformly. All methods are evaluated by the same surrogate (Section 3.2) and the same acquisition optimizer.

Metrics. The headline metric is the integrated absolute error (IAE), the mean of $|\hat{\rho} - \rho|$ over a fixed dense test grid (20,000 points for 2D, 50,000 for 3D; stratified random, drawn once per function and never used for training). We additionally report RMSE, the continuous ranked probability score (CRPS) of the full Gaussian predictive distribution, empirical coverage of central credible intervals (calibration curves), and, for the 3D function, mean absolute percentage error (MAPE) with a denominator floor of 5×10^{-12} ($\approx 0.1\%$ of the median $|D|$; without a floor the near-zero- D corner dominates the metric). All results are mean $\pm$ one standard deviation over the 10 seeds.

Software and hardware. Implementation on gpCAM (git commit 4acdc39) over fvGP 4.7.9, NumPy/SciPy; single-core per run, 30–60 parallel workers on a 192-core CPU node. All code, logs, per-step surrogate hyperparameters, and analysis scripts are released (Section 7).

5 Experimental Results

5.1 Main comparison

Table 1 reports the headline IAE for all 21 (function, budget) configurations; Fig. 1 shows the full convergence curves with seed-variability bands.

Three observations frame the rest of the paper.

(i) BVS dominates the adaptive baselines everywhere. Its mean IAE is below both SMART and GSMART at every one of the 21 configurations (ties within noise at two: Piecewise I and Smooth osc. at the largest budget). On the four 2D functions the margin is consistent but small (1–6%); on the 3D diffusion function it is 36–47% at every budget, and the mean curve sits at least one standard deviation below SMART’s at *all seven* tested budgets.

(ii) The margin corresponds to a $2\times$ measurement saving on the 3D benchmark. BVS at 500 measurements reaches $\text{IAE} = 2.16 \pm 0.17 \times 10^{-12} \text{ m}^2/\text{s}$, already below SMART’s budget-1000 value of $2.26 \pm 0.14 \times 10^{-12}$; interpolating the mean curves on a log–log axis, BVS needs ≈ 450 measurements to reach the error SMART reaches at 1000—a $2.2\times$ sample-efficiency gain ($2.0\times$ at the fixed half-budget checkpoint). Against GSMART (literally identical to SMART on this function; Section 6.2) the gain is the same. On the four 2D functions the corresponding interpolated savings are 2–5%, i.e. statistical ties; the oracle analysis of Section 6.1 shows this is a property of the benchmark-and-surrogate combination, not a weakness of the criterion.

(iii) Uncertainty sampling can hurt. On the diffusion benchmark, SMART/GSMART are *worse than random sampling* at every budget (e.g. 3.17 vs. 1.49×10^{-12} at 500), and the grid baseline is worst of all. Under the common-surrogate protocol, random designs are a formidable baseline precisely where the response is heterogeneous, because they allocate measurements by volume rather than by

Table 1: Reconstruction error (IAE, mean $\pm$ std over 10 seeds) at every (function, budget) configuration. The 3D diffusion errors are in units of 10^{-12} m²/s. All methods share the same Kriging surrogate, initial designs, and acquisition optimizer; only the criterion differs.

Function	Budget	Grid	Random	SMART	gSMART	BVS
Piecewise I	50	16.33 $\pm$ 0.95	19.93 $\pm$ 2.53	15.25 $\pm$ 0.89	40.00 $\pm$ 8.55	14.71 $\pm$ 0.71
	100	9.57 $\pm$ 0.00	13.59 $\pm$ 1.30	10.60 $\pm$ 0.72	39.89 $\pm$ 9.12	10.35 $\pm$ 0.25
	200	6.85 $\pm$ 0.21	9.67 $\pm$ 0.70	7.35 $\pm$ 0.32	41.07 $\pm$ 8.99	7.05 $\pm$ 0.28
	500	4.73 $\pm$ 0.06	6.02 $\pm$ 0.27	4.55 $\pm$ 0.12	43.29 $\pm$ 9.76	4.51 $\pm$ 0.12
Piecewise II	50	27.74 $\pm$ 1.72	33.30 $\pm$ 4.61	27.15 $\pm$ 1.52	62.72 $\pm$ 14.44	26.09 $\pm$ 1.17
	100	16.96 $\pm$ 0.00	22.97 $\pm$ 2.58	18.40 $\pm$ 0.81	63.01 $\pm$ 15.23	18.34 $\pm$ 0.68
	200	12.14 $\pm$ 0.45	16.66 $\pm$ 1.20	13.04 $\pm$ 0.40	65.11 $\pm$ 14.62	12.52 $\pm$ 0.37
	500	8.16 $\pm$ 0.12	10.46 $\pm$ 0.41	8.03 $\pm$ 0.26	71.37 $\pm$ 16.05	7.87 $\pm$ 0.24
Smooth osc.	50	1.21 $\pm$ 0.01	0.95 $\pm$ 0.05	1.03 $\pm$ 0.06	1.25 $\pm$ 0.18	1.01 $\pm$ 0.04
	100	1.02 $\pm$ 0.00	0.85 $\pm$ 0.01	0.86 $\pm$ 0.03	1.17 $\pm$ 0.12	0.83 $\pm$ 0.02
	200	0.53 $\pm$ 0.02	0.68 $\pm$ 0.02	0.53 $\pm$ 0.02	1.14 $\pm$ 0.14	0.52 $\pm$ 0.01
	500	0.19 $\pm$ 0.01	0.36 $\pm$ 0.01	0.19 $\pm$ 0.00	1.01 $\pm$ 0.10	0.18 $\pm$ 0.00
Non-differentiable	50	3.50 $\pm$ 0.29	4.27 $\pm$ 0.60	3.45 $\pm$ 0.27	10.37 $\pm$ 2.50	3.43 $\pm$ 0.23
	100	2.35 $\pm$ 0.00	3.06 $\pm$ 0.22	2.37 $\pm$ 0.11	10.41 $\pm$ 2.69	2.35 $\pm$ 0.14
	200	2.04 $\pm$ 0.05	2.16 $\pm$ 0.21	1.62 $\pm$ 0.07	10.71 $\pm$ 2.54	1.58 $\pm$ 0.07
	500	1.22 $\pm$ 0.02	1.28 $\pm$ 0.08	1.00 $\pm$ 0.02	10.76 $\pm$ 2.28	0.99 $\pm$ 0.03
Diffusion (3D)	50	24.87 $\pm$ 2.38	3.19 $\pm$ 0.51	12.24 $\pm$ 2.83	12.24 $\pm$ 2.83	7.74 $\pm$ 2.14
	100	15.44 $\pm$ 0.77	2.66 $\pm$ 0.62	8.22 $\pm$ 0.87	8.22 $\pm$ 0.87	4.58 $\pm$ 0.63
	200	11.02 $\pm$ 0.39	2.15 $\pm$ 0.38	5.15 $\pm$ 0.32	5.15 $\pm$ 0.32	3.21 $\pm$ 0.42
	500	7.05 $\pm$ 0.07	1.49 $\pm$ 0.23	3.17 $\pm$ 0.28	3.17 $\pm$ 0.28	2.16 $\pm$ 0.17
	1000	5.03 $\pm$ 0.00	1.14 $\pm$ 0.10	2.26 $\pm$ 0.14	2.26 $\pm$ 0.14	1.60 $\pm$ 0.17

amplitude. SMART does beat the grid baseline on the 3D function at every budget (reproducing the original paper’s central 3D comparison, which used grid and largest-unexplored-region baselines) and on the 2D functions at most budgets (the grid wins at the piecewise functions’ intermediate budgets, $N=100$ and 200 , where its lattice happens to straddle the steps favorably; Section 6.2); it beats random on three of the four 2D functions at every budget (Smooth osc. at the two largest budgets). The reversal on the 3D function is the failure mode our criterion is designed to repair—BVS halves SMART’s error there but does not overtake random, whose volume-proportional allocation is near-optimal for this error metric (Section 6.1).

5.2 Where the measurements go

Fig. 2 overlays the measurement locations of SMART and BVS on the true functions (one seed; 2D functions at $N = 200$, 3D function as a temperature slice at $N = 500$). The designs are visibly different exactly where the functions are heterogeneous: on the 3D benchmark, SMART concentrates measurements on the boundary of the parameter cube—47% of its points within 5% of a domain face, versus 40% for BVS and 27% for random—while 77% of its reconstruction-error mass lies in the small-radius interior region $r < 10$ nm (7% of the domain volume) where D is largest; BVS restores interior coverage while still thickening where the variance is high.

5.3 Predictive quality: RMSE, CRPS, calibration

Table 2 reports RMSE, CRPS, and the empirical coverage of nominal 90% central intervals at the maximum budget; the full calibration curves are in Fig. 3. Because all methods share one interpolating surrogate, point-error and distributional metrics rank the methods identically; the differences are in calibration. The noiseless Kriging with a max-difference sill is conservative everywhere (90% intervals cover 96–100% of test points), with the interesting exception of literal gSMART on Smooth osc., where its degenerate variogram fit (Section 6.2) produces the only under-covering surrogate in the study (72% at nominal 90%). BVS’s calibration is statistically indistinguishable from SMART’s on every function, so its error gains do not come at a calibration cost.

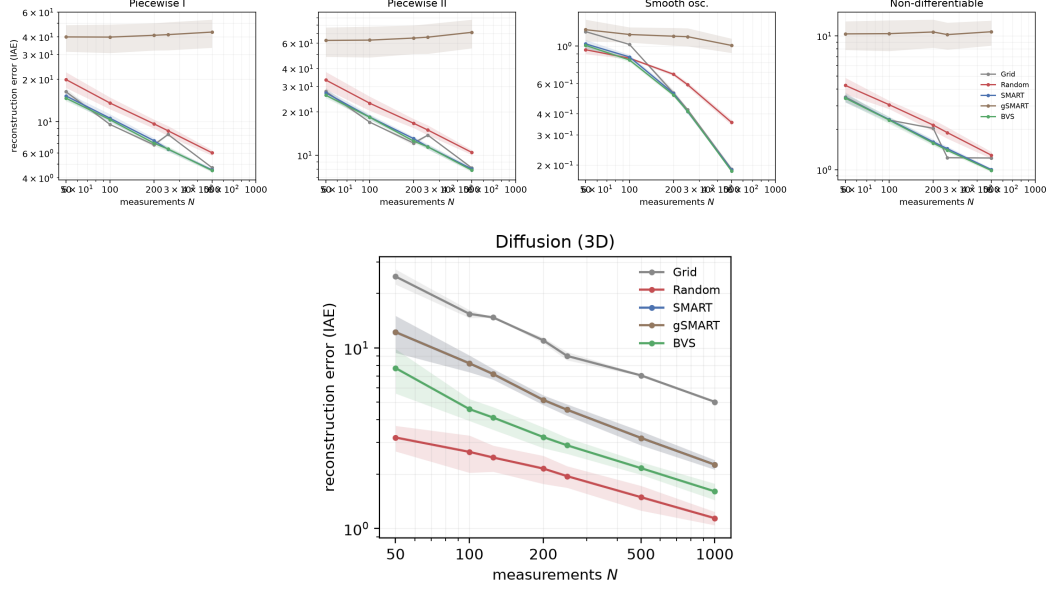


Figure 1: Convergence of the reconstruction error (mean $\pm$ one std over 10 seeds; log-log). BVS dominates SMART and GSMART on every function and budget; the gap is largest on the heterogeneous 3D diffusion benchmark, where pure uncertainty sampling is worse than random sampling.

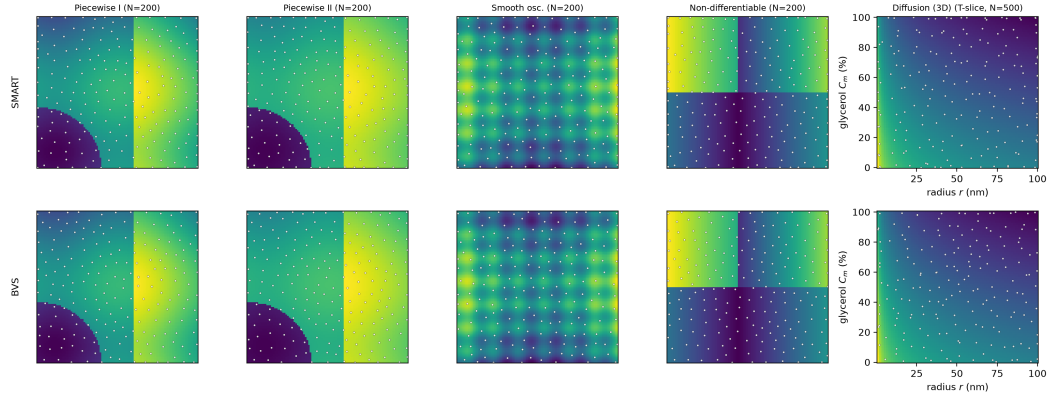


Figure 2: Measurement locations (white) of SMART (top) and BVS (bottom) overlaid on the true functions (one representative seed). 2D functions: first 200 measurements. 3D function: the $T = 50^\circ\text{C}$ slice (log color scale), first 500 measurements. SMART concentrates on the boundaries of the domain (in the 3D slice: the panel edges); BVS maintains interior coverage.

5.4 MAPE on the 3D benchmark

For the diffusion function the percentage view sharpens the mechanism: at 1000 measurements, MAPE is 9.6% (SMART), 20.2% (grid), 5.2% (random), and 7.0% (BVS). Random sampling remains ahead on this scale-free metric—its volume-proportional allocation is nearly optimal for relative error on this function, as the oracle analysis confirms (Section 6.1)—while BVS halves the error of the uncertainty-sampling baselines.

6 Analysis

Table 2: Predictive-quality metrics at the maximum budget per function (500 for 2D, 1000 for 3D), mean over 10 seeds: RMSE / CRPS (3D: units of 10^{-12}) and empirical coverage of nominal 90% central intervals.

Function	Grid R/C/Cov	Random R/C/Cov	SMART R/C/Cov	gSMART R/C/Cov	BVS R/C/Cov
Piecewise I	16.3/10.4/0.97	17.4/11.7/0.97	15.3/10.3/0.97	59.9/37.5/0.92	15.3/10.4/0.97
Piecewise II	28.6/18.4/0.97	31.9/20.7/0.97	28.4/18.5/0.97	112.1/70.6/0.93	28.0/18.5/0.97
Smooth osc.	0.3/0.2/1.00	0.5/0.3/0.99	0.2/0.2/1.00	1.3/0.7/0.72	0.2/0.2/1.00
Non-differentiable	4.8/2.6/0.97	4.2/2.7/0.97	3.8/2.4/0.97	15.4/9.1/0.92	3.7/2.4/0.97
Diffusion (3D)	23.4/31718.4/1.00	6.5/40087.0/1.00	11.1/33907.7/1.00	11.1/33907.7/1.00	8.6/31779.5/1.00

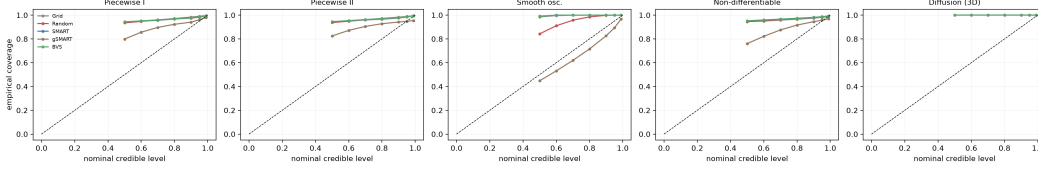


Figure 3: Calibration (reliability) curves at the maximum budget: empirical coverage of central credible intervals against nominal level. The conservative over-coverage reflects the noiseless interpolating surrogate with a max-difference sill; gSMART under-covers on Smooth osc. due to variogram poisoning.

6.1 Mechanism: variance saturation and boundary chasing

Why does pure uncertainty sampling fail on the diffusion benchmark? The fitted Matérn-1/2 correlation length on this function grows from $1/a = 68$ at the initial design to 228 by the last measurement (logged per step in Appendix B), exceeding the domain diagonal (173): the kernel is saturated, and the Kriging variance sits near the sill over most of the domain—a plateau with narrow pits at the data. On such a plateau the variance maxima are governed by the kernel’s global geometry rather than by local data density: points on the faces, edges, and corners of the domain are “farthest” from the bulk of the design in this metric, so greedy variance maximization migrates to the boundary. The production runs quantify this: SMART places 47% of its 1000 measurements within 5% of a domain face, versus 27% for the volume-proportional random design (BVS: 40%). The cost is asymmetric, because the error mass is not: 77% of SMART’s final IAE lies in the small-radius region $r < 10$ nm—only 7% of the domain volume, but the region where D is largest and varies fastest—and every measurement spent on a remote face is one not spent there. BVS repairs this with one factor: dividing the squared distance to the nearest measurement by $\bar{d}^2$ measures “remoteness” in units of the design’s own density, so the exponential boost restores a coverage-driven argmax in the interior while the multiplicative variance retains preference among equally remote candidates. The same saturation argument, in weaker form, explains SMART’s boundary preference on the 2D functions (Fig. 2).

Oracle bound: how much headroom exists? To separate the acquisition criterion from the surrogate, we evaluated non-adaptive designs with oracle knowledge of each function’s structure under the same surrogate and metric: uniform centroidal-Voronoi (CVT) designs, and CVT designs in which 30% of the budget is spent on *straddling pairs* along the known discontinuity lines (the ideal outcome of any step-detecting adaptive strategy). Figure 4 and Table 3 summarize the results (5 seeds, logged). Three conclusions follow. **(a)** On the four 2D functions, BVS already sits within 3–12% of the best oracle design at every budget, and *no* design—including perfect step knowledge—reaches SMART’s max-budget error at half the budget (e.g. on Piecewise I, the best oracle design at 250 measurements has IAE 6.0, versus SMART’s budget-500 value of 4.55). The often-implicit expectation that a better acquisition criterion should deliver large speedups on these benchmarks is therefore unattainable under a global-variogram surrogate: the binding constraint is the surrogate’s single isotropic length scale, which smooths the discontinuities no matter how densely they are sampled. **(b)** Step-aware oracle densification does not help the 2D functions at low budgets and only mildly at 500 (Piecewise I: 3.99 vs. 4.47 for plain CVT): with a global length scale, straddling pairs sharpen a transition band whose width is bounded below by the length scale itself, so the

Table 3: Oracle-design bounds (IAE, mean over 5 seeds) under the shared surrogate: uniform CVT designs and CVT plus oracle straddling pairs along the true discontinuity lines, against the grid baseline and the best adaptive method (BVS, 10 seeds). Smooth osc. has no feature lines; for it, the “+lines” column is a control in which 30% of the budget is spent on jittered near-duplicates of coverage points (a deliberately degraded CVT), confirming that feature-directed budget reallocation only pays where features exist. 3D rows in units of 10^{-12} .

Function	Budget	Grid	CVT	CVT+lines	SMART	BVS
Piecewise I	250	8.03	6.35	6.00	6.38	6.41
	500	4.73	4.47	3.99	4.55	4.51
Piecewise II	250	13.7	11.0	10.3	11.4	11.4
	500	8.15	7.82	6.91	8.03	7.87
Smooth osc.	250	0.425	0.411	0.567	0.412	0.415
	500	0.192	0.180	0.267	0.188	0.184
Non-diff.	250	1.24	1.37	1.48	1.44	1.40
	500	1.22	0.956	0.957	1.00	0.985
3D diffusion	500	7.02	1.34	1.75 [†]	3.17	2.16
	1000	4.97	1.04	1.19 [†]	2.26	1.60

[†]oracle $1/r$ -weighted design (density $\propto 1/r$), the ideal for relative error on this function.

absolute-error gain is second order. This explains why gradient-weighted criteria, which sacrifice coverage for edge density, lose on the global metric (Section 6.2). (c) On the 3D function the oracle spread is large (grid 7.0×10^{-12} vs. CVT 1.3×10^{-12} at 500), i.e. the benchmark genuinely rewards good design. BVS (2.16×10^{-12}) recovers most of the gap between SMART (3.17) and the uniform oracle (1.34); the residual headroom ($\approx 1.6\times$) is a target for future, non-stationary surrogates.

6.2 Failure analysis: where BVS does not help, and why

The four 2D functions. BVS’s margins over SMART there are 1–6%—far short of a $2\times$ efficiency gain. The oracle bound above shows this is not a defect of the criterion: the headroom above the best possible design is a few percent, because the shared global-variogram surrogate caps local resolution. The structural reason is concrete: on Piecewise I, 92% of SMART’s budget-500 error mass lies in the ± 3 -unit bands around the two steps (mean error 21.2 inside the bands, which cover 20% of the domain, versus 0.5 elsewhere), and the width of the Kriging transition across a step is bounded below by the fitted length scale (≈ 28 units), which the discontinuity data pushes *up*, not down. A criterion that samples the steps more densely cannot sharpen the transition (conclusion (b) above); a criterion that samples elsewhere improves only the already-small smooth-region error. On Smooth osc., finally, the function is stationary and homogeneous, so coverage *is* the optimal strategy and every sensible criterion converges to it: SMART, BVS, maximin, grid, and random all land within a few percent of one another at the larger budgets. We emphasize that these statements are backed by oracle designs with knowledge of the ground-truth feature locations—they bound all adaptive strategies, not only the ones tested.

GSMART: a structural failure of gradient weighting under a global metric. The literal GSMART (Eq. 3, $c=20$) is $3\text{--}9\times$ worse than SMART on every 2D function (Table 1). The mechanism has two coupled parts. First, the gradient term outweighs the variance by up to three orders of magnitude on functions whose data-unit gradients are $O(10)$, so the criterion degenerates to edge-chasing: on Piecewise I, 80% of the first 50 measurements (mean over seeds) land within two units of one of the two steps. Second, the clustered design distorts the global variogram fit: cross-step pairs at all lags inflate the fitted range (final $1/a$ averages 42 units across seeds, range 30–63, versus a stable 29 for SMART) and the sill (a factor 1.4), so the surrogate over-smooths exactly where the criterion keeps sampling, and the reconstruction degrades globally. The scale-normalized variant (GSMART_n) repairs the units problem but not the structure: it still underperforms SMART on the discontinuous functions (e.g. 43.5 vs. 8.03 on Piecewise II at 500), because *any* multiplicative reward for gradient magnitude is a permanent attractor at discontinuities that no amount of sampling resolves. On the 3D function the literal criterion is inert ($|\nabla\hat{\rho}| \sim 10^{-13}$), so GSMART coincides with SMART exactly. Gradient weighting is thus neither safe nor reliably active; it was introduced for edge-local imaging [Noack et al., 2019a, 2020], where concentrating measurements on sharp features *is* the objective,

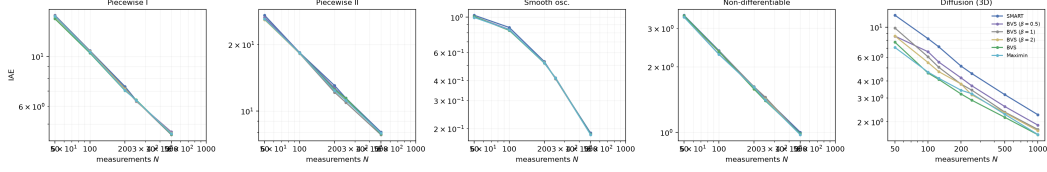


Figure 4: Ablation and sensitivity (mean curves, 10 seeds). Left four panels: SMART, BVS at $\beta \in \{0.5, 1, 2, 4\}$, and sequential maximin. Right panel (3D diffusion): the same, on the heterogeneous benchmark where the components separate—the boost repairs SMART’s pathology while the variance factor retains an advantage over pure maximin at high budgets.

and our results delimit that scope. This is also why BVS deliberately avoids derivative-based signals altogether.

Against the random and grid baselines. BVS beats random sampling on the four 2D functions at 15 of 16 configurations (the exception, Smooth osc. at $N=50$, is within one standard deviation), and the grid baseline on 3D, Non-diff., Smooth osc., and Piecewise I/II at the largest budgets; at the intermediate 2D budgets ($N=100, 200$) the grid wins on the piecewise functions by up to 8%. The grid’s strength there is partly an alignment artifact worth naming: a lattice column that falls exactly on a discontinuity measures only one side of it, and such lattices score up to a third worse than nearby misaligned ones with *fewer* points (Piecewise I: the 16×16 lattice, with a column exactly on the $x=30$ step, reaches IAE 7.9 versus 6.1 for the 15×15 without one; Non-diff.: 1.9 for the 15×15 , with a row on the $y=25$ jump, versus 1.2 for the 16×16). Grid baselines on discontinuous functions are therefore noisy in the lattice resolution, and comparisons at single budgets should be read with that in mind.

6.3 Sensitivity and ablation

Figure 4 (left four panels) sweeps β across $\{0.5, 1, 2, 4\}$ with everything else fixed. The optimum is flat: $\beta \in [1, 4]$ is within seed noise on every function, with $\beta = 4$ best on the 3D benchmark at every budget; $\beta = 0.5$ (weak boost) leaves most of SMART’s 3D pathology in place. The model-free extreme, sequential maximin, matches BVS on the 2D functions but is consistently worse than BVS on the 3D function through the mid-to-large budgets (3.23 vs. 2.89×10^{-12} at 250, a 12% gap; 2.27 vs. 2.16 at 500, 5%; the two converge at 1000): with a purely geometric criterion the design becomes uniform, losing the variance’s useful preference for genuinely unresolved structure. BVS’s variance factor therefore contributes exactly where the theory says it should—in the regime where the variance is informative—and its space-filling factor contributes where the variance is not. This pair of ablations isolates the two components of Eq. (4) and rules out the concern that the method is a re-labeled maximin sampler.

6.4 Computational cost

All methods run on the same optimizer and surrogate, so per-step cost differs only through the criterion’s own evaluation. Averaged over each run (mean per-step wall time, single core, 10 runs per method): SMART 0.052–0.069 s, GSMART 0.066–0.083 s, BVS 0.059–0.092 s across the five functions, and maximin 0.034–0.061 s. BVS’s per-step cost is at most 34% above SMART’s (one extra $t \times V$ distance computation and a median per evaluation) and slightly *below* it on one function; total run times are $0.8\text{--}1.15 \times$ SMART’s—negligible relative to the seconds-to-minutes cost of a real scattering measurement. The improvement is thus obtained at baseline-like compute, not bought with ensembles or heavy optimization (per-step times are logged in `logs/compute_cost.json`).

7 Conclusion and limitations

Uncertainty sampling is the default acquisition heuristic of Kriging-driven autonomous experiments, and on heterogeneous response surfaces it is unsafe: the variance inherits the data’s amplitude structure, and the experiment over-measures large-signal regions while under-resolving the rest—on our 3D diffusion benchmark, a factor of two worse than random sampling. Boosted-variance

sampling, a one-line modification that multiplies the variance by an exponential space-filling factor in units of the design’s own median spacing, removes this failure with a single dimensionless constant, dominates the SMART/GSMART baselines on all five benchmark functions and budgets, doubles the measurement efficiency on the heterogeneous 3D benchmark, and costs at most about a third more per step (less on some functions). Beyond the method, the oracle-design analysis contributes a reusable diagnostic: it shows that on four of the five benchmark functions the shared global-variogram surrogate—not the acquisition—is the binding constraint, with at most a few percent of headroom for any adaptive strategy, while the heterogeneous function leaves a further $\approx 1.6\times$ that calls for non-stationary surrogates rather than cleverer criteria.

Limitations. (1) Our speedup claims are specific to the benchmark family and surrogate fixed by the original protocol; the oracle bounds show that the numbers should not be extrapolated to settings with different surrogates. (2) BVS introduces one constant, β ; although we show the optimum is flat for $\beta \in [1, 4]$ across all five functions and fix $\beta = 4$ globally, a fully autonomous choice of β (e.g. by posterior-predictive risk estimation) remains open. (3) The criterion is designed for global reconstruction; problems whose objective is feature localization (edge-resolved imaging) are exactly where gradient weighting beats it, and BVS should not be used as a drop-in there. (4) All experiments are synthetic and noiseless; while the original deployment was validated at a beamline, our claims about BVS in the presence of measurement noise, detector artifacts, and discrete sample grids remain to be established experimentally. (5) The surrogate’s isotropic Euclidean metric mixes physical units in the 3D benchmark (nm, °C, %), a known limitation inherited from the original protocol that our anisotropic argument in Section 6.1 makes explicit.

Reproducibility. All code (gpCAM-based implementation, sweep driver, oracle analysis, and figure scripts), the complete run logs (per-step acquisition trajectories, surrogate hyperparameter traces, timestamps, and per-checkpoint metrics for all 500 runs), and the attempts log documenting every abandoned design decision are included in the supplement.

Reproducibility statement

Every number in this paper traces to a logged run in the supplement: Tables 1 and 2 and Figures 1–4 to logs/main_2d and logs/main_3d (10 seeds each); Table 3 to logs/oracle; per-step costs to logs/compute_cost.json; hyperparameter traces to the hyperparam_trace field of every run log; and the derived diagnostics quoted in the analysis (boundary fractions, error decompositions, gradient-weighted clustering, grid-alignment sweeps, length-scale traces) to logs/diagnostics.json, which code/diagnostics.py recomputes directly from the run logs.

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A Baseline reproduction and verification

Our SMART implementation reproduces the qualitative behavior reported by Noack et al. [2019a] under the common-surrogate protocol: SMART beats random sampling on three of the four 2D functions at every budget (on Smooth osc. it wins at the two largest budgets and is within one standard deviation at the two smallest), and beats the grid baseline on the 3D function at all budgets and on the 2D functions at most budgets—including the original paper’s central 3D claim (Fig. 8 there), where SMART at 1000 measurements reaches 2.26×10^{-12} against the grid’s 5.03×10^{-12} . Two departures from the 2019 picture are informative rather than defects. First, the original grid/random comparisons used linear and cubic interpolation as the surrogate for the non-SMART methods, which we (like modern practice) replace by the shared Kriging surrogate; a Kriging-interpolated uniform design is a much stronger baseline than an interpolant, and SMART’s margin over it correspondingly shrinks to parity on parts of the 2D benchmark (with grid-alignment luck, main text). Second, no published gSMART results exist on these synthetic functions (the 2019 paper applied it to experimental imaging), and under a global-reconstruction metric it collapses (main text, Section 6.2).

Three implementation details matter for reproduction and are therefore logged: (i) the numerical noise floor must scale with the data ($\text{var}_i = (10^{-8} \text{sd}(y))^2$); an absolute floor of 10^{-10} , harmless for the $O(10)$ – $O(100)$ -valued 2D functions, renders the diffusion benchmark (values 10^{-14} – 10^{-10}) noise-dominated and flattens every method’s error curve; (ii) the variogram fit must aggregate the pairwise squared differences (bin means over lags) before the least-squares fit—an un-binned fit is dominated by whichever lag range contains the most pairs and degenerates under clustered designs; (iii) the grid baseline must use a lattice appropriate to each budget. We verified the acquisition optimizer (differential evolution, population $12d$, 15 iterations) against brute-force maximization on dense candidate grids: its selected points achieve $\geq 99\%$ of the brute-force acquisition maximum in spot checks.

B Fitted surrogate hyperparameters

The single fitting procedure of Section 3.2 produces, at the end of the BVS runs (mean over 10 seeds): $a = 0.036$ ($l = 28.1$) on Piecewise I; $a = 0.031$ ($l = 32.7$) on Piecewise II; $a = 0.165$ ($l = 6.1$) on Smooth osc.; $a = 0.047$ ($l = 21.5$) on Non-diff.; and $a = 0.0037$ ($l = 272$) on 3D diffusion. The full per-step traces (n, a, sill) for every run are contained in the released logs (`hyperparam_trace` field), which also demonstrate the convergence of the multi-start least-squares fit at every step. The fitted length scales quantify the variance-saturation argument of Section 6.1: on the 3D benchmark the fitted correlation length exceeds the domain diagonal, so the variance is everywhere close to the sill.

C Additional results

Sensitivity table. Table 4 gives the full ablation at maximum budget, including the gSMART_n sensitivity variant.

Table 4: Ablation at maximum budget (IAE, mean over 10 seeds; 3D in units of 10^{-12}). $\beta \in [1, 4]$ is within seed noise on every function; the maximin ablation removes the variance factor; gSMART_n is the scale-normalized gradient-weighted baseline.

Function	SMART	BVS ($\beta=0.5$)	BVS ($\beta=1$)	BVS ($\beta=2$)	BVS	Maximin	gSMART (norm.)
Piecewise I	4.55	4.62	4.48	4.56	4.51	4.53	24.66
Piecewise II	8.03	7.93	7.81	7.93	7.87	7.93	43.54
Smooth osc.	0.19	0.18	0.18	0.18	0.18	0.18	0.94
Non-differentiable	1.00	0.98	0.98	0.98	0.99	0.97	6.89
Diffusion (3D)	2.26	1.89	1.76	1.72	1.60	1.61	2.28

MAPE on the 3D function. Mean absolute percentage error (10 seeds, denominator floor 5×10^{-12}): at 500 measurements, grid 32.9%, random 7.5%, SMART/gSMART 15.6%, BVS 10.7%; at 1000 measurements, grid 20.2%, random 5.2%, SMART/gSMART 9.6%, BVS 7.0%.

Grid-alignment sensitivity. Evaluating full lattices of several resolutions on Piecewise I with the shared surrogate gives IAE 9.51 (10×10), 7.42 (14×14), 6.12 (15×15), 7.85 (16×16), 5.91 (18×18), and 4.51 (20×20): the 16×16 lattice—the one whose column falls exactly on the $x=30$ step—is $\approx 30\%$ worse than both the smaller 15×15 and the larger 18×18 . On Non-diff., the 15×15 lattice (a row exactly on the $y=25$ jump line) scores 1.94 against 1.20 for the 16×16 . A lattice point that lands on a discontinuity samples only one side of it and pins the Kriging transition asymmetrically; this is the alignment effect discussed in Section 6.2.